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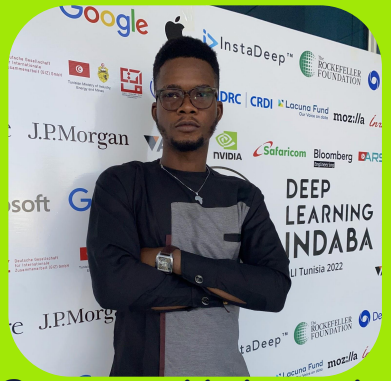
Building and Enlightening Data Professionals in Africa.

An annual conference for all data practitioners in Africa.

#DataFestAfrica22
#DFA22

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The path to ML mastery and it's not “what you are thinking.”



Steven Kolawole

ML Collective | Nazari

@steveddev

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“What you are thinking”...

- Take Python, data science, and machine learning courses and, maybe math classes too.
- Do Kaggle and Zindi competitions to build your machine learning skills.
- Write technical articles. Try to teach others.
- Learn software engineering for model deployment and DevOps for performance monitoring.
- Build personal projects. Maybe build an E2E ML application incorporating data engineering and MLOps.
- Contribute to open-source.

It's Research!




There are dozens of us!

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Profile Summary

- ML Researcher @ ML Collective
- ML Engineer @ Nazari | ASAlytics

- Low-budget philosopher
- Clowning-as-a-Service



Steven Kolawole

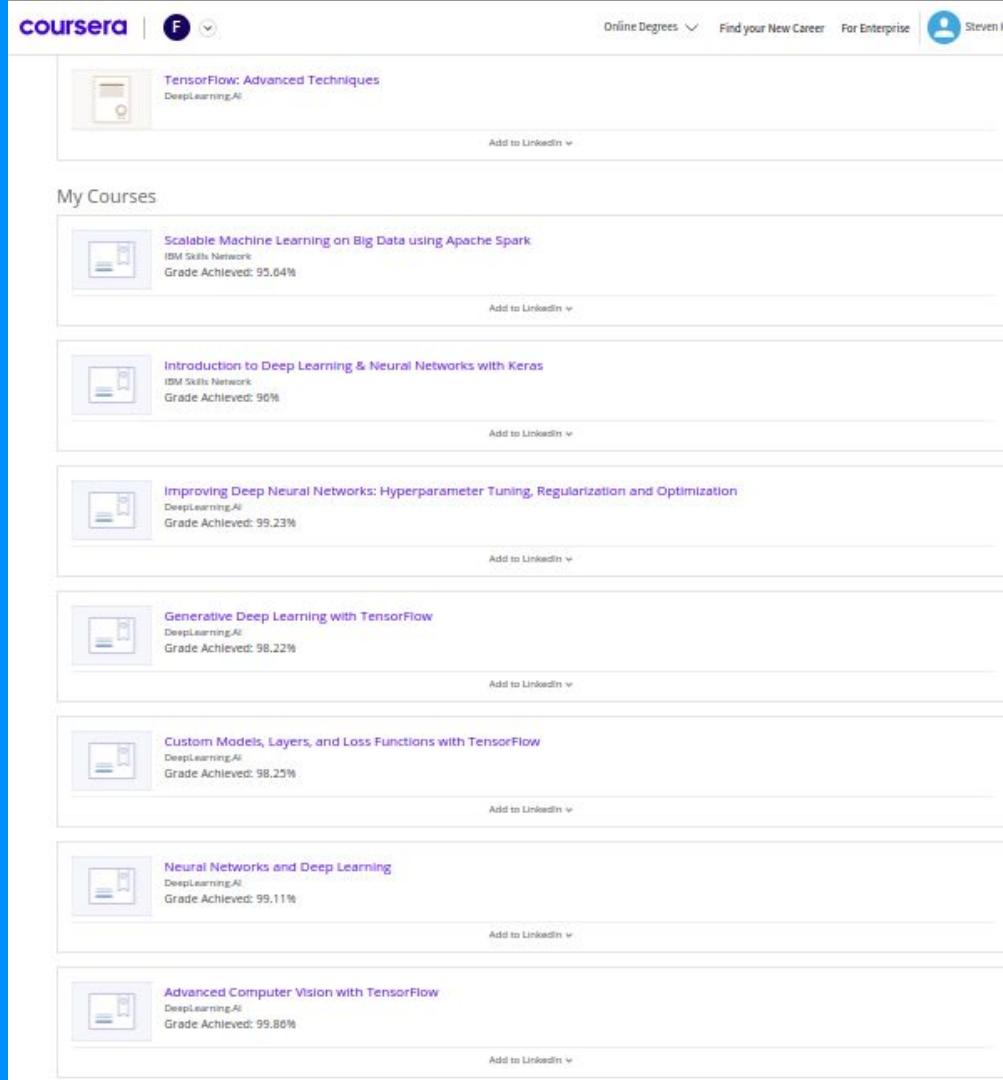
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I know a few things about Python.



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I know a thing or two about taking Python/ML/Math s courses.



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I know a few things about building ML projects, open source dev, and winning hackathons.





nigeria computer society
ncs
www.ncs.org.ng

March 11, 2022

Mr. Steven Kolawole
27, Ajewole St. Oke-Aregba
Abeokuta, Ogun State.

Dear Mr. Kolawole,

NCS ARTIFICIAL INTELLIGENCE SUMMIT 2022: AI CHALLENGE WINNER

Greetings from the Nigeria Computer Society (NCS), the umbrella body for all Information Technology Professionals, Interest Groups, and Stakeholders.

The National Executive Council (NEC) and the entire membership of NCS wish to congratulate you for coming out at the **First Position** in the AI Challenge that took place during the just-concluded Artificial Intelligence Summit held at Lafia, Nasarawa State.

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I know a few things about ML Research.

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Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language

Steven Kolawole[1*], Opeyemi Osakuade[2], Nayan Saxena[1] and Babatunde Kazeem Olorisade[3]
[1]ML Collective
[2]Data Science Nigeria
[3]Cardiff Metropolitan University

Evolutionary Federated Learning Using Particle Swarm Optimization

OPT2022: 14th Annual Workshop on Optimization for Machine Learning

Vision Transformer for Edge Devices - A Short Overview

sites.google.com/view/broadening-collaboration-in-ml/speakers?authuser=0

Home | Call for papers | **Speakers** | Schedule | Organizers | Program comittee

McGill University, DeepMind — Learn more

Professor, Universidad Nacional de Córdoba — Learn more

Masters student, MILA — Learn more



Kanjun Qiu

CEO, Generally Intelligent

Learn more



Steven Kolawole

Independent Researcher, ML Collective

Learn more



Jesse Dodge

Allen Institute for AI

Learn more

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What I assume about you



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A couple of Caveats

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A couple of Caveats

- I am not a top 1% in all the paths in “what you’re thinking“ so some of my hot takes might be wrong.

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A couple of Caveats

- I am not a top 1% in all the paths in “what you’re thinking“ so some of my hot takes might be wrong.
- It’s not possible that all my hot takes would be wrong.

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“What you’re thinking” #1:

Taking courses

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Why I like courses

- A very good way to start out your ML journey.
- Helps you hone specific skills in more advanced topics.
- Instils standard practice - super-useful when just starting out.
- Courses help you gain momentum.

DATAQUEST

udemy

freeCodeCamp

sololearn

coursera

PLURALSIGHT

IBM Developer SKILLS NETWORK

datacamp

codecademy

edX

UDACITY

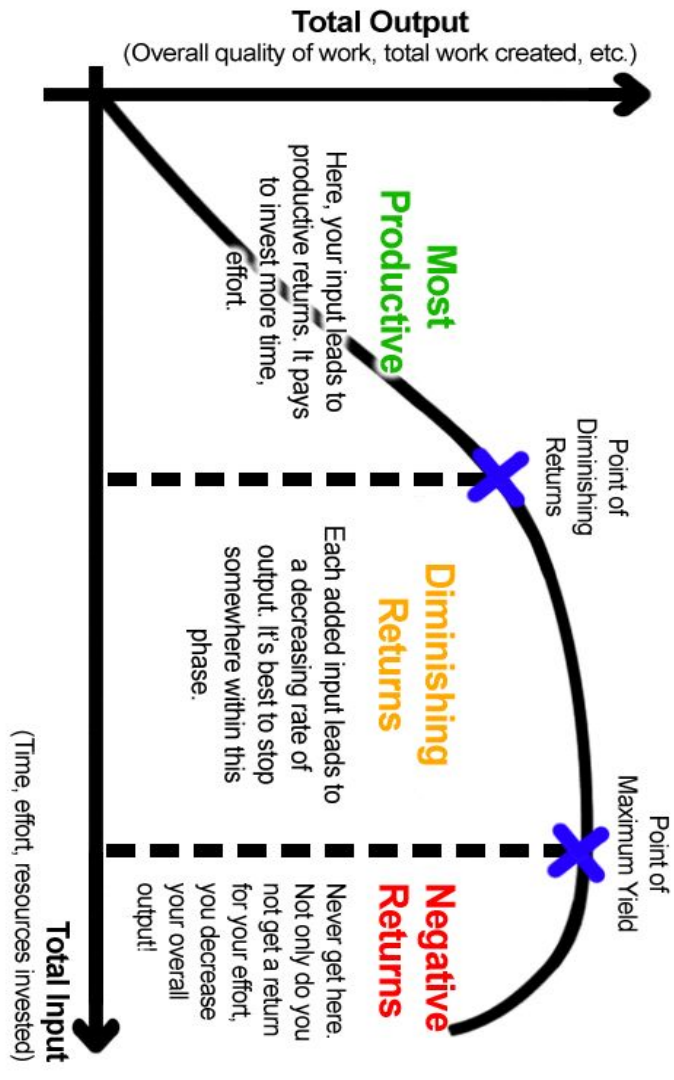
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But...

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Momentum will stop momentum-ing 🤌🏾

Tutorial Hell is real and it's one of humanity's worst known sufferings. You can easily find lots of noobs burning, out there, in the fire of tutorials.





The law of Diminishing Returns

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- ML Courses are usually high-level. Great for newbies, not great for non-newbies.
- The low-level ML courses are usually too “boring” or “dense”.
- Huge gap between courses and real-life.
- Courses, like meetings, create an illusion of being super resourceful.
- You’d probably forget most of what you learned in that course in 6 months. Or less.

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Can go without saying but...

You most likely won't be able to function in a junior ML role after finishing the most popular ML course in the world.



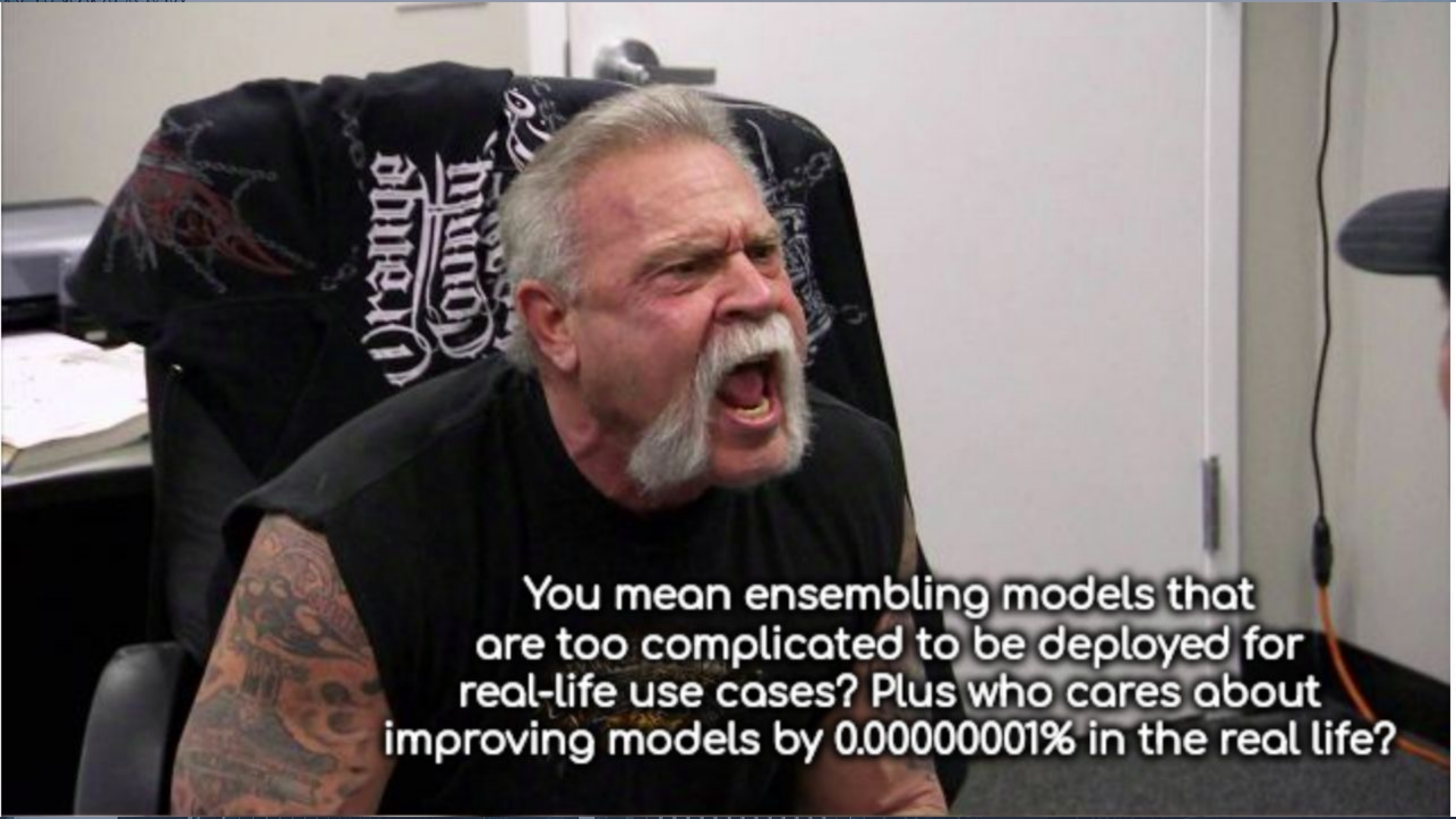
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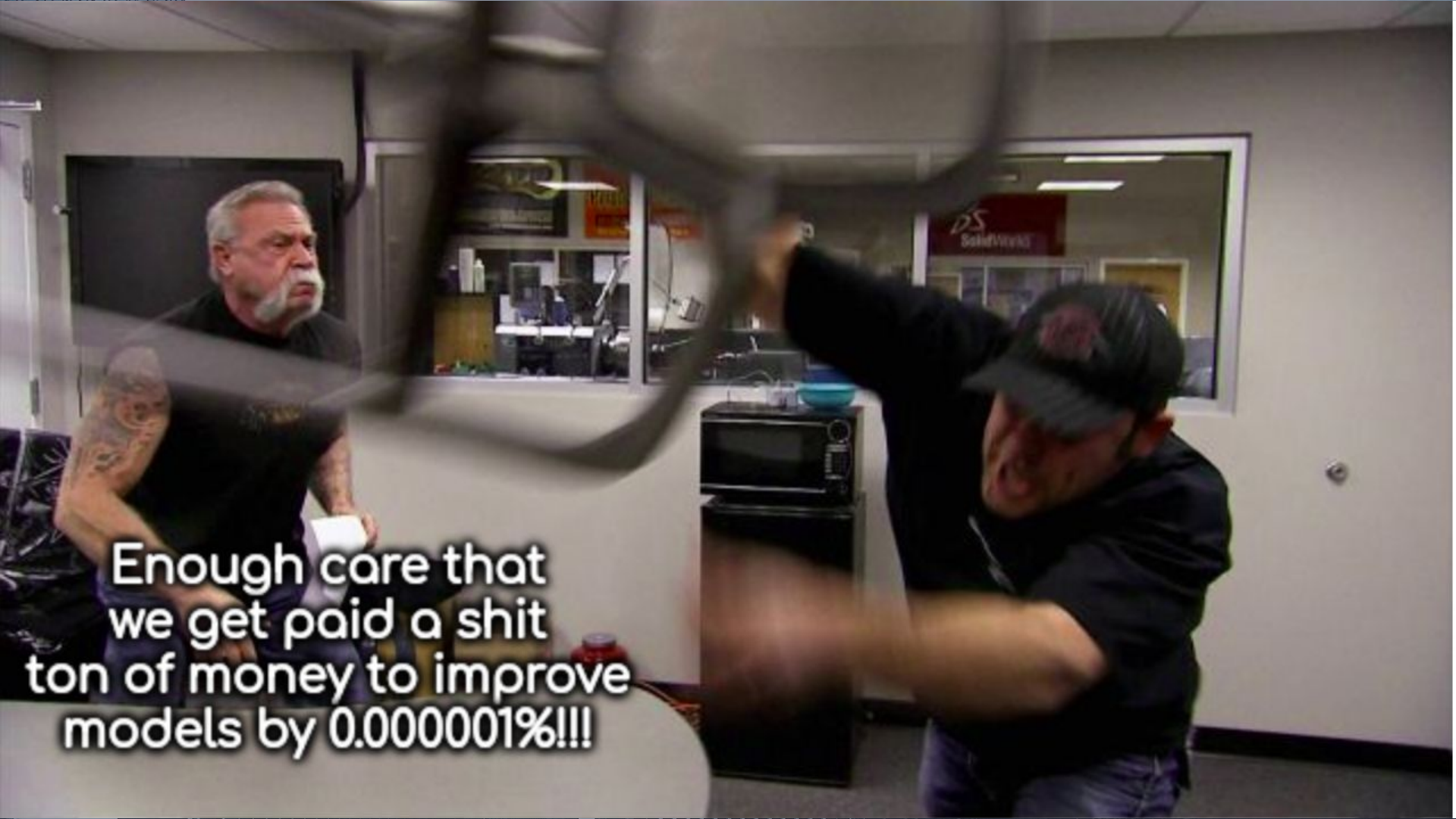
“What you’re thinking” #2:

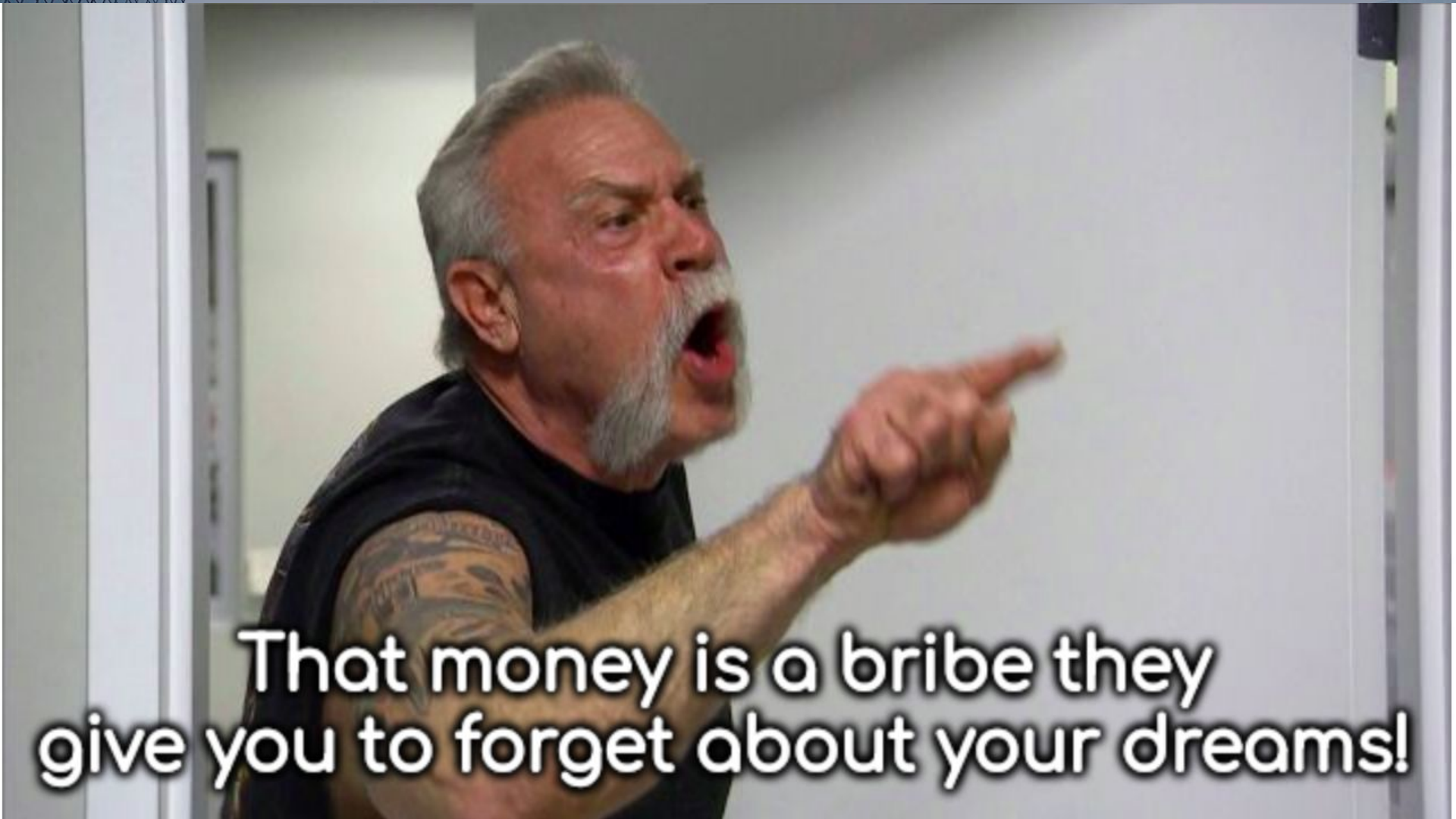
Participating in Kaggle/Zindi Competitions


Your Kaggle/Zindi Competitions are
like porn, you know. They are not real.


But we do Data cleaning, EDA
- looks pretty real-life to me.
And we engineer features and ensemble
models like you've never seen before!


You mean ensembling models that
are too complicated to be deployed for
real-life use cases? Plus who cares about
improving models by 0.000000001% in the real life?


Enough care that
we get paid a shit
ton of money to improve
models by 0.000001%!!!


That money is a bribe they
give you to forget about your dreams!

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“What you’re thinking” #3:

Write technical articles. You become a master by teaching others.

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By teaching we learn, but,



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By teaching we learn, but,

- It is super easy to get stuck in the reward loop and forget the reason for towing this path in the first place.



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“What you’re thinking” #4, #5 & #6:

Spend time on Software Engineering

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The many pros of SWE in your ML trajectory

- Your models are not just ending up in IPYNB files,
- You become a better & well-rounded programmer and versed in standard practices as a result,
- Building E2E ML apps will always stand out on most CVs,
- Contributing to open-source will feel like working in a top-tier firm, all points checked,
- If you work in an AI/ML startup, SWE will be important as you'd need to be a generalist lots of time,
- etc.

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But...

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- Tendency for knowledge base to increase in breadth while lacking depth.

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ML Interviews:

Tendency for knowledge base to increase in breadth while lacking depth.



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- Tendency for knowledge base to increase in breadth while lacking depth.
- Tendency to switch to SWE completely.

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Tendency to switch to SWE completely.



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So why Research?

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- Keep up with the state-of-the-art in machine learning.



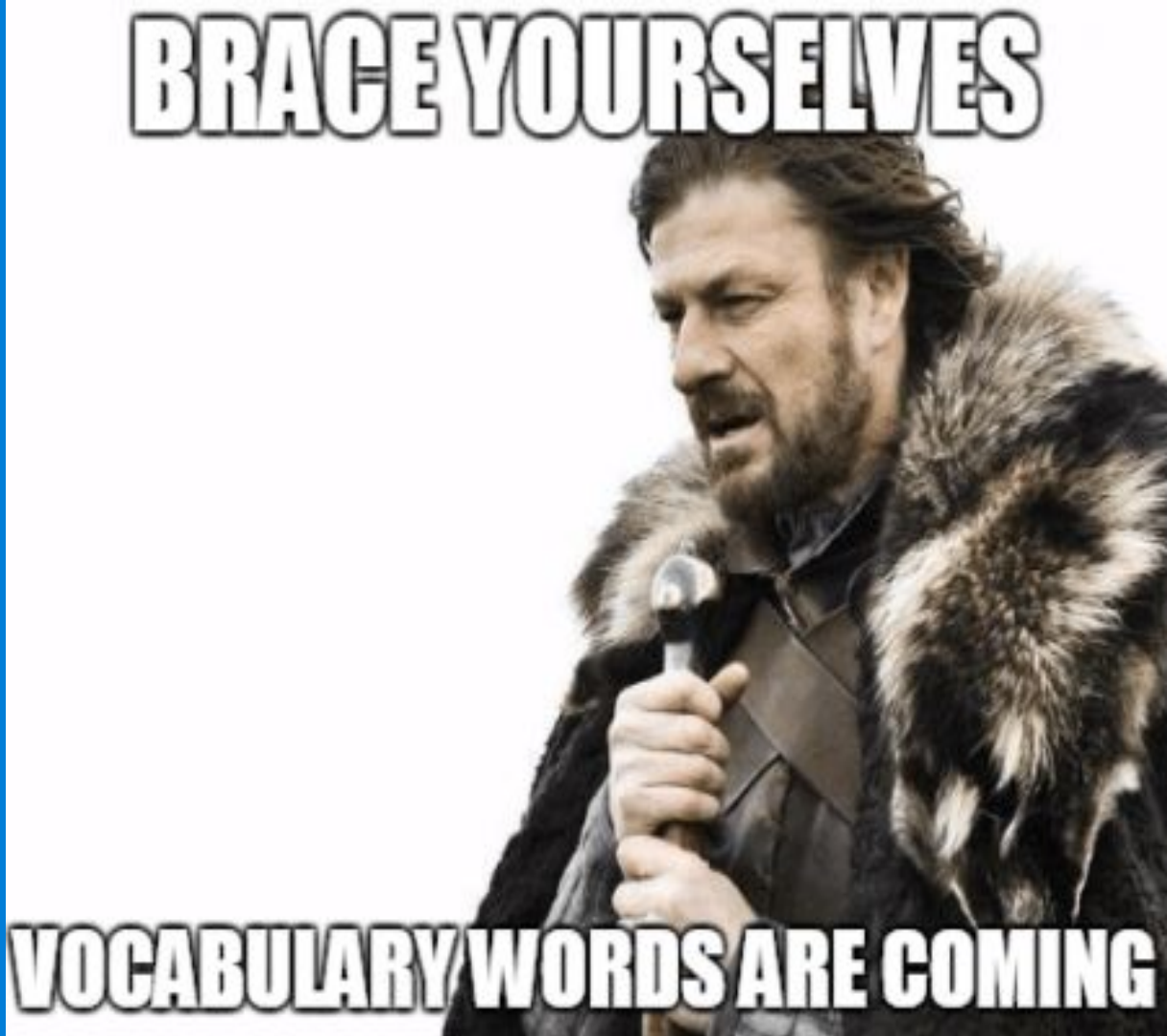
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- Understand the building blocks of the ML tools you use regularly.



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- Your interests in ML is extremely limited without exploring research.



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- **Reproducible science:** Open source documentation and standard SWE practice.



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- You might enjoy research enough to transition from an ML Engineer to a Research Engineer.



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In conclusion...

- You don't need to be sophisticated with ML before you start reading/implementing research.

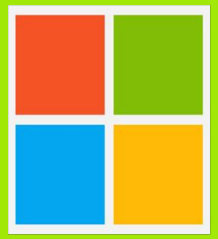
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In conclusion...

- You don't need to be sophisticated with ML before you start reading/implementing research.
- *No dulling, chiefs!* Start now!!!

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Thank you Sponsors

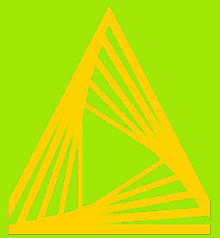













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THANK YOU!



Steven Kolawole

ML Collective | Nazari

@steveddev

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